



ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Data Science

CERTIFICATE

This is to certify that the Cornerstone project work entitled “**ATTENDANCE DATASET ANALYSIS**” is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Information Technology for the academic year 2025-2026.

Project Guide

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DECLARATION

We hereby declare that the Cornerstone project entitled “**ATTENDANCE DATASET ANALYSIS**” is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree in Data Science**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

Attendance Analysis System is a system developed to examine and evaluate the impact of student attendance on academic achievement. In many educational institutions, attendance data is collected regularly but is not utilized effectively for analytical purposes. This project aims to transform raw attendance data into meaningful insights by applying data mining techniques.

The system considers key attributes such as attendance percentage, internal assessment marks, final examination marks, and CGPA. These attributes are analyzed using machine learning algorithms to identify patterns and relationships between attendance and performance. Data preprocessing is performed to clean and prepare the dataset by removing duplicates, handling missing values, and normalizing data to improve analysis accuracy.

The J48 Decision Tree algorithm is used for classification, which categorizes students into performance groups such as Good, Average, and Poor. Additionally, the K-Means clustering algorithm is used to group students based on similarities in their academic behavior.

Visualization techniques such as scatter plots are used to clearly represent the relationship between attendance and academic performance. The results obtained from the analysis show that students with higher attendance tend to perform better academically, while students with low attendance are more likely to have poor performance.

This system helps in early identification of weak students and supports educators in taking corrective measures such as mentoring and academic support. Overall, the project demonstrates how data mining techniques can be effectively applied in the educational domain to improve student performance and decision-making.

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ACRONYMS

- **CGPA** : Cumulative Grade Point Average
- **WEKA** : Waikato Environment for Knowledge Analysis
- **ML** : Machine Learning
- **CSV** : Comma Separated Values
- **J48** : Decision Tree Algorithm

1. INTRODUCTION

1.1 Brief Information about the Project

The Attendance Analysis System is a data mining application designed to analyze student attendance data and its impact on academic performance. Attendance is a critical factor in the learning process, as regular participation in classes enhances understanding and improves academic outcomes.

In traditional systems, attendance is mainly recorded for administrative purposes and is rarely used for analysis. However, with the advancement of data mining techniques, it is possible to extract valuable insights from attendance data. This project uses machine learning algorithms to analyze the relationship between attendance and performance, helping educators make informed decisions.

The system processes the dataset using the WEKA tool, which provides various features for data preprocessing, classification, clustering, visualization and association rule mining. By analyzing attendance data, the system helps identify patterns that indicate how attendance affects student performance.

1.2 Motivation and Contribution of Project

The motivation behind this project arises from the lack of effective utilization of student data in educational institutions. Although large amounts of attendance and performance data are collected, they are not analyzed to improve student outcomes.

One of the major problems in traditional systems is the late identification of weak students. By the time poor performance is noticed, it becomes difficult to take corrective action. This project aims to solve this problem by analyzing data at an early stage.

The contributions of this project include:

- Providing a data-driven approach to analyze student performance
- Identifying weak students early
- Reducing manual effort in analyzing data
- Improving accuracy and reliability of analysis
- Supporting educators in decision-making

1.3 Objectives of the Project

The primary objective of this project is to analyze student attendance data and understand its impact on academic performance. The system aims to use data mining techniques to extract meaningful insights from the dataset.

The specific objectives include:

- To classify students into performance categories
- To apply clustering techniques for grouping students
- To analyze the relationship between attendance and performance
- To visualize data for better understanding
- To provide insights that help improve academic results

1.4 Scope of the Project

The scope of the Attendance Analysis System includes analyzing student attendance data and applying data mining techniques to evaluate academic performance. The system can be used in schools, colleges, and universities to monitor student progress.

It provides a simple and effective way to identify weak students and understand performance patterns. The project can be extended to include additional features such as predictive analysis, real-time monitoring, and integration with academic systems.

2. SOFTWARE ANALYSIS

2.1 Software Requirements

The Attendance Analysis System requires a suitable software environment for efficient implementation and execution of data mining tasks. The primary tool used in this project is WEKA (Waikato Environment for Knowledge Analysis), which is an opensource software designed for machine learning and data analysis. WEKA provides the users to perform various operations such as data preprocessing, classification, clustering, and visualization without requiring advanced programming knowledge. The system operates on a Windows operating system, which ensures compatibility and smooth performance of the WEKA tool.

The dataset used in this project is stored in CSV (Comma Separated Values) format, which is easy to manage and widely supported by WEKA. Basic tools like Microsoft Excel or any text editor can be used to create, view, and modify the dataset. The system requires a minimum of 4GB RAM to handle the dataset efficiently and ensure smooth processing during analysis. Internet connectivity is needed for downloading the dataset and necessary software components. The WEKA Explorer interface is used to carry out all data mining operations, including data cleaning, normalization, and attribute selection. It supports various algorithms such as J48 for classification and KMeans for clustering, which are essential for this project.

Additionally, WEKA provides built-in visualization tools that help in representing data graphically, making it easier to understand patterns and relationships. The software environment is simple, flexible, and does not require high-end hardware, making it suitable for academic purposes. Overall, the software requirements are minimal yet sufficient to successfully implement and analyze the Attendance Analysis System.

3. TECHNOLOGY DESCRIPTION

3.1 Techniques Used

WEKA (Waikato Environment for Knowledge Analysis)

WEKA (Waikato Environment for Knowledge Analysis) is an open-source machine learning tool developed by the University of Waikato. It is widely used for data mining, data analysis, and predictive modeling. WEKA provides a rich set of algorithms for tasks such as classification, clustering, regression, and association rule mining. It supports multiple data formats like CSV and ARFF, making dataset handling simple and flexible. The tool offers a user-friendly graphical interface, which allows users to perform complex operations without deep programming knowledge. This makes it highly suitable for students, researchers, and beginners in machine learning. In this project, WEKA is used to analyze student attendance and academic performance data.

The dataset is first preprocessed using techniques like data cleaning, handling missing values, normalization, and attribute selection. These steps improve the quality and accuracy of the data. For classification, the J48 Decision Tree algorithm is used to categorize students into groups such as Good, Average, and Poor.

For clustering, the K-Means algorithm groups students based on similarities in their data. Association rule mining using the Apriori algorithm helps identify patterns, such as the relationship between attendance and marks. WEKA also provides visualization tools like graphs and scatter plots to better understand the data.

Overall, WEKA is an efficient and reliable tool for applying data mining techniques and extracting meaningful insights from educational datasets.



Figure 3.1: WEKA Tool

3.2 METHODS USED

3.2.1 Data Preprocessing

Data preprocessing is one of the most important steps in data mining, as the quality of input data directly affects the accuracy of the results. Raw data collected from realworld sources often contains noise, missing values, duplicate entries, and irrelevant attributes. Therefore, preprocessing is required to clean and prepare the dataset before applying machine learning algorithms.

In this project, several preprocessing techniques are applied. Duplicate records are removed to avoid redundancy and incorrect analysis. Missing values are identified and handled appropriately to ensure completeness of the dataset. Irrelevant attributes that do not contribute to performance analysis are removed to reduce complexity.

Normalization is applied to bring all numerical values into a common range, which helps in improving the performance of algorithms. Standardization is also used to ensure that data follows a consistent distribution. These preprocessing steps improve data quality, making the dataset suitable for accurate and reliable analysis.

3.2.2 Classification

Classification is a supervised learning technique used to categorize data into predefined classes based on input features. In this project, classification is used to predict student performance based on attendance and marks.

The J48 Decision Tree algorithm is used for classification. It is an extension of the ID3 algorithm and is widely used due to its simplicity and effectiveness. The algorithm builds a tree structure where each internal node represents a decision based on an attribute, and each leaf node represents a class label.

In this system, attributes such as attendance percentage, internal marks, and final marks are used to classify students into categories such as Good, Average, and Poor. The decision tree helps in understanding how different attributes influence student performance. It also provides a clear and interpretable model for prediction.

3.2.3 Clustering

Clustering is an unsupervised learning technique used to group data points based on similarity without predefined labels. In this project, clustering is used to identify groups of students with similar academic characteristics.

The K-Means algorithm is used for clustering. It works by dividing the dataset into a predefined number of clusters. Each cluster is represented by a centroid, and data points are assigned to the cluster with the nearest centroid.

In this system, students are grouped based on attributes such as attendance and marks. This helps in identifying patterns such as groups of high-performing students and groups of low-performing students. Clustering provides additional insights that are not obtained through classification alone.

3.2.4 Visualization

Visualization is the process of representing data in graphical form to make it easier to understand patterns and relationships. In this project, visualization is used to analyze the relationship between attendance and academic performance.

Scatter plots are used to display the correlation between attendance percentage and CGPA. Each point in the graph represents a student, making it easy to observe trends.

Visualization helps in validating the results obtained from classification and clustering.

It also improves the overall presentation of the project by making complex data easy to interpret. Graphical representations are essential for effective communication of results.

3.2.5 Association Rule Mining

Association rule mining is a data mining technique used to discover relationships between different attributes in a dataset. In this project, it is used to identify patterns between attendance, marks, and student performance.

The Apriori algorithm is applied to generate rules based on support and confidence values. These rules help in identifying meaningful relationships, such as low attendance leading to poor performance or high attendance resulting in good performance.

This technique helps in uncovering hidden patterns in the dataset and provides additional insights that support better decision-making in academic analysis.

4. DATASET DESCRIPTION

4.1 Student Dataset

The dataset used in this project is obtained from Kaggle, which is a well-known platform for data science and machine learning datasets. The dataset contains approximately 2500 records, where each record represents an individual student.

The data is stored in CSV (Comma Separated Values) format, which is easy to import and process using the WEKA tool. The dataset includes important attributes such as attendance percentage, internal marks, final examination marks, and CGPA.

The dataset is structured and organized, making it suitable for data mining operations. It provides sufficient information to analyze student performance and identify patterns. The use of a real-world dataset enhances the reliability and relevance of the project.

4.2 Dataset Attributes

The dataset used in this project contains detailed information related to student attendance and academic performance. Each attribute plays an important role in analyzing the relationship between attendance and student results.

Table 4.1: Dataset Attributes Description

Attribute Name	Data Type	Description
Student ID	Numeric	Unique identifier for each student
Class	Categorical	Class or year of the student
Section	Categorical	Section of the student (A/B/C)
Subject	Categorical	Subject name
Total Classes	Numeric	Total number of classes conducted
Classes Attended	Numeric	Number of classes attended by the student
Attendance Percentage	Numeric	Percentage of attendance
Internal Marks	Numeric	Marks obtained in internal examinations
Final Exam Marks	Numeric	Marks obtained in final examination
CGPA	Numeric	Overall academic performance

Performance Status	Categorical	Student performance category (Good/Average/Poor)
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The above attributes are selected to provide a comprehensive analysis of student performance. Attendance-related attributes such as total classes, classes attended, and attendance percentage help in measuring student regularity. Academic attributes such as internal marks, final exam marks, and CGPA represent the performance level of students. The **Performance Status** attribute is used as the target variable for classification. By using these attributes, the system can analyze patterns, identify weak students, and understand the impact of attendance on academic performance.

5. MODULES DESCRIPTION

5.1 Data Collection Module

The Data Collection Module is responsible for gathering and loading the dataset into the system. The dataset is collected from Kaggle and initially contained 2505 records, out of which 5 duplicate records were removed, resulting in 2500 final records. The dataset is stored in CSV format and includes 11 attributes such as Attendance Percentage, Internal Marks, Final Exam Marks, CGPA, and Performance Status. This module ensures proper integration of the dataset into the WEKA tool for further analysis.

5.2 Preprocessing Module

The Data Preprocessing Module focuses on cleaning and preparing the dataset to improve its quality. First, duplicate records are removed to avoid redundancy and incorrect analysis. Next, irrelevant attributes such as Student ID, Total Classes, and Classes Attended are removed as they do not contribute to prediction. The dataset is then checked for missing values, and it is observed that no missing values are present. After cleaning, normalization is applied to scale all numerical values between 0 and 1, and standardization is performed to transform data with a mean of 0 and standard deviation of 1. These steps ensure accurate and efficient data analysis.

5.3 Classification Module

The Classification Module is used to categorize students into performance groups such as Good, Average, and Poor. The J48 Decision Tree algorithm is applied using attributes like Attendance Percentage and Internal Marks. The model achieved an accuracy of 79.72%, indicating reliable performance. The confusion matrix shows correct predictions of 1139 Average, 650 Poor, and 204 Good instances. This module helps in early identification of weak students and provides a clear understanding of factors affecting performance.

5.4 Clustering Module

The Clustering Module groups students based on similarities in their academic data using the K-Means algorithm. The dataset contains 2500 instances, which are divided into 2 clusters after 8 iterations.

- Cluster 0: 1417 students (57%) – Average performance
- Cluster 1: 1083 students (43%) – Poor performance

The clustering is performed using attributes such as Attendance, Internal Marks, Final Exam Marks, and CGPA. This module helps in identifying hidden patterns and grouping students based on performance levels.

5.5 Visualization Module

The Visualization Module presents the analysis results in graphical form for better understanding. A scatter plot is used to show the relationship between Attendance Percentage and CGPA. The visualization clearly indicates that students with higher attendance tend to have higher CGPA, while students with lower attendance show poor academic performance. This module helps in easily identifying trends and validating the results obtained from classification and clustering.

5.6 Association Rule Mining Module

The Association Rule Mining Module is used to identify relationships between different attributes in the dataset. The Apriori algorithm is applied to generate rules based on minimum support of 0.02 and minimum confidence of 0.6. A total of 20 rules are generated using 2500 instances, with frequent itemsets such as L(1): 33, L(2): 180, and L(3): 58. The analysis shows that low attendance is strongly associated with poor performance, while high attendance leads to good performance. This module helps in discovering hidden patterns and supports better decision-making.

6. SYSTEM IMPLEMENTATION

The system implementation is carried out using the WEKA data mining tool. The implementation process involves loading the dataset, performing preprocessing, applying classification and clustering algorithms, and visualizing the results. Each step is executed in the WEKA Explorer interface and is explained below with corresponding figures

6.1 Loading the Dataset

The dataset in CSV format is loaded into WEKA using the “Open File” option in the Explorer. The attributes such as attendance percentage, internal marks, final marks, and CGPA are displayed.

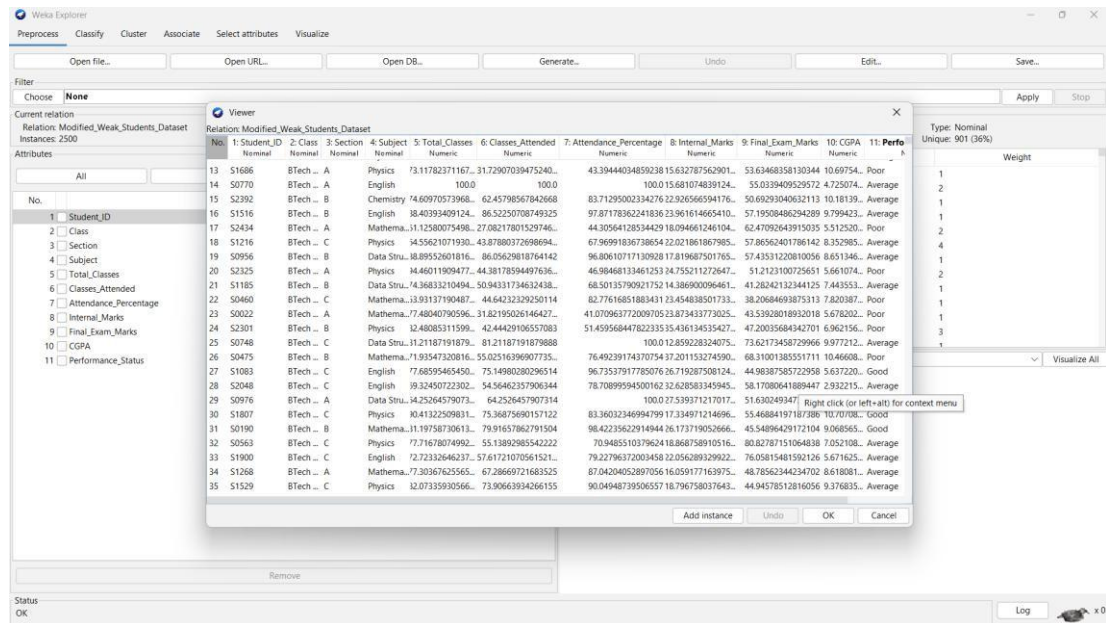


Figure 6.1: Dataset Loaded in WEKA

6.2 Data Preprocessing

In this step, preprocessing techniques are applied using the “Preprocess” tab in WEKA. Missing values are handled, duplicate entries are removed, and unnecessary attributes are filtered. Normalization is also applied to scale the data.

6.2.1 Duplicate Removal

This figure shows the process of identifying duplicate records in the dataset. Duplicate entries are removed to avoid redundancy and incorrect analysis. This improves the accuracy and reliability of the dataset

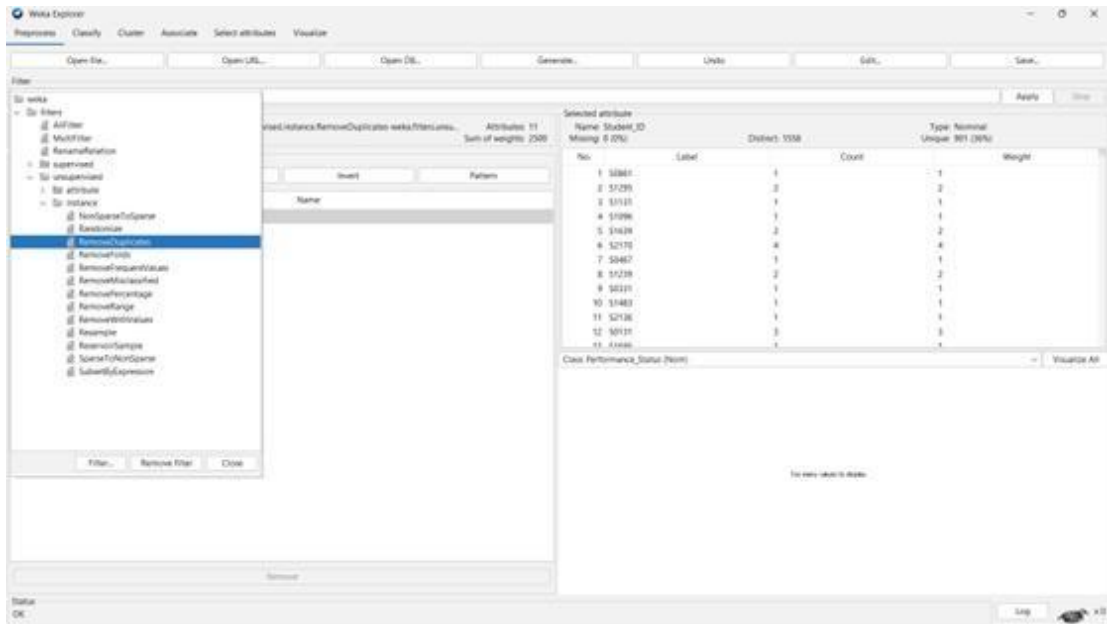


Figure 6.2: Duplicate Removal in WEKA

6.2.2 Irrelevant Attribute Removal

This figure shows the removal of irrelevant attributes from the dataset. Only important features related to attendance and marks are retained. This helps in improving model efficiency and reducing complexity.

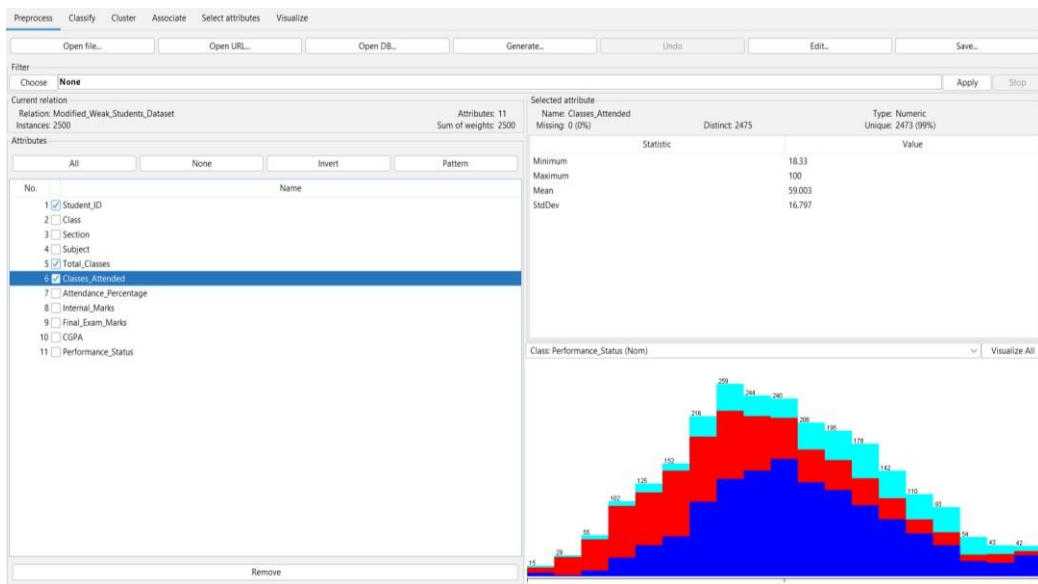


Figure 6.3: Attribute Selection in WEKA

6.2.3 Missing Value Check

This figure represents the detection of missing values in the dataset. Missing data is handled by removing or replacing values. This ensures completeness and consistency of the dataset..

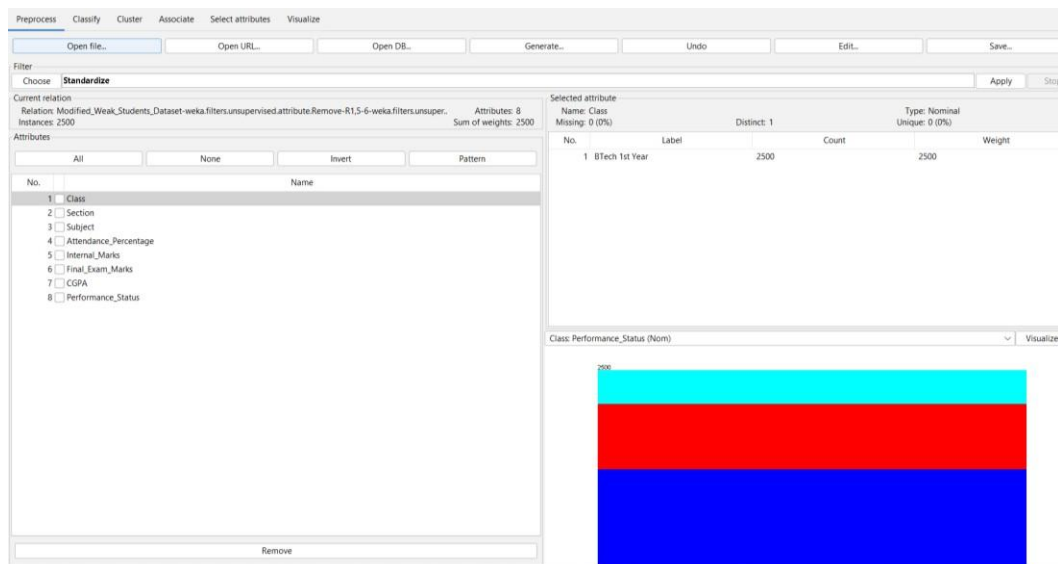


Figure 6.4: Missing Value Handling

6.2.4 Normalization

This figure shows the normalization of numerical data. Values are scaled into a common range between 0 and 1. This improves the performance of machine learning algorithms.

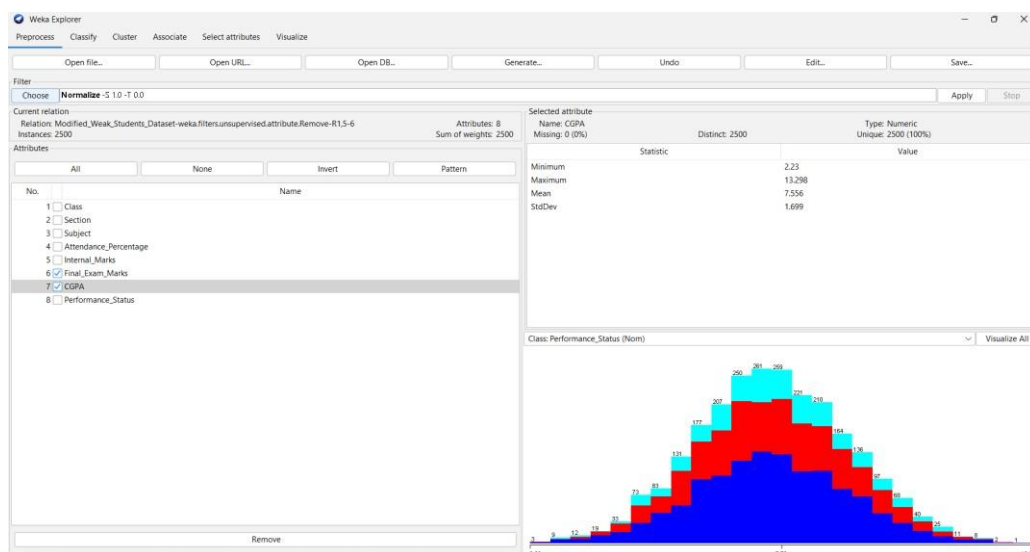


Figure 6.9: Normalization Process

6.2.5 Standardization

This figure represents the standardization of data distribution. Data is transformed to have a mean of zero and a standard deviation of one. This helps in improving clustering and classification results.

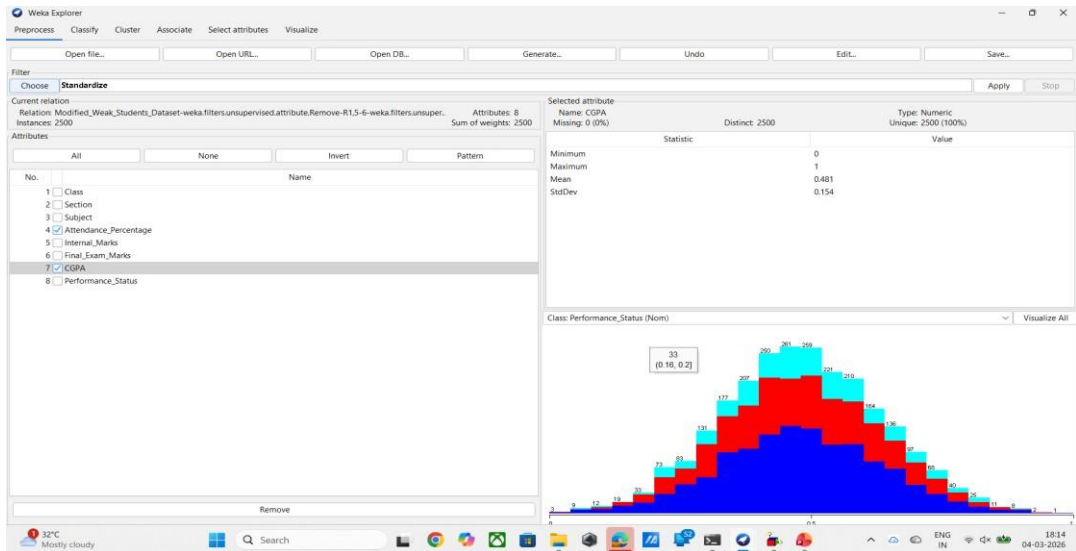


Figure 6.10: Standardization Process

6.3 Classification using J48 Algorithm

This figure shows the output of the J48 classification algorithm. Students are classified into Good, Average, and Poor categories. It helps in predicting student performance.

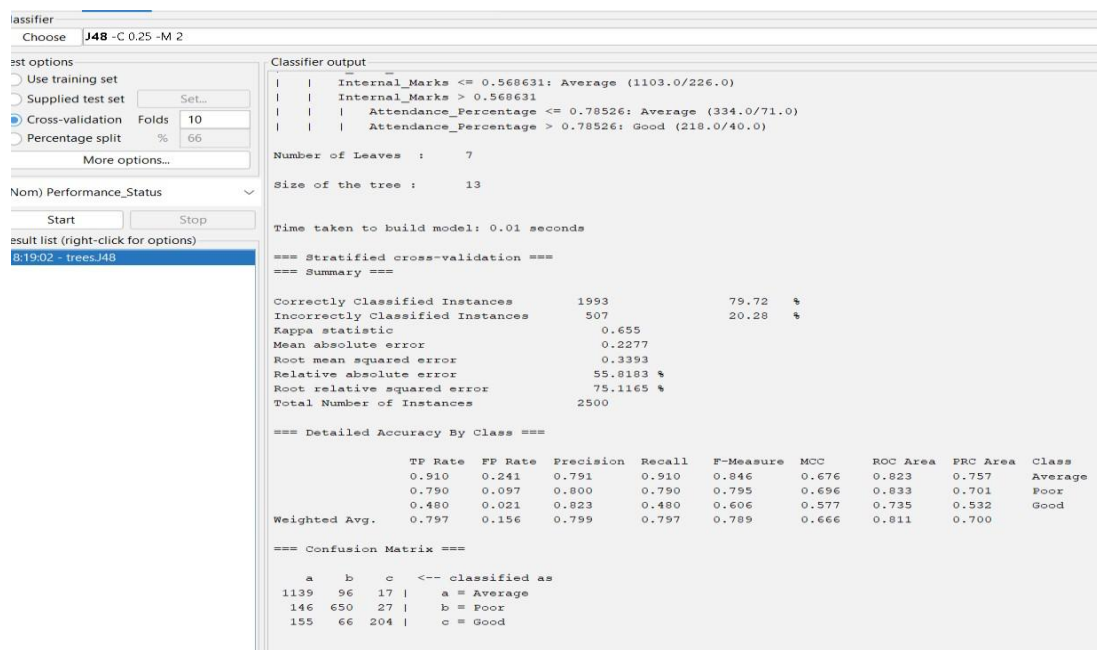


Figure 6.11: J48 Classification Output

6.3.1 Confusion Matrix

This figure shows the confusion matrix of the classification model. It represents correct and incorrect predictions for each class. This helps in evaluating the accuracy of the model.

=== Confusion Matrix ===

a	b	c	<-- classified as
1139	56	17	a = Average
146	650	27	b = Poor
155	66	204	c = Good

Figure 6.12: Confusion Matrix for J48 Classification

6.3.2 J48 Decision Tree Output

This figure shows the decision tree generated by the J48 algorithm. It represents decision rules based on attendance and marks. It helps in understanding how classification is performed.

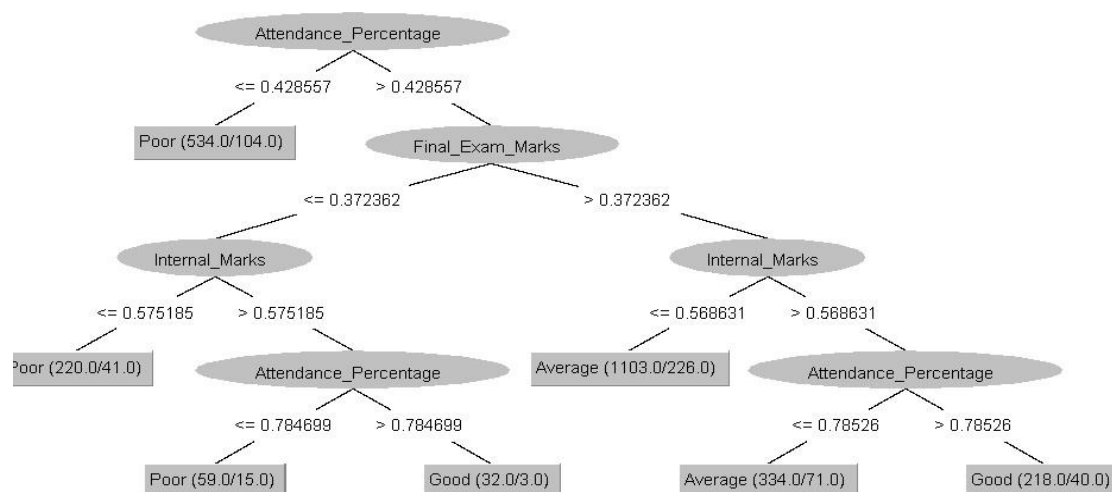


Figure 6.13: J48 Decision Tree Output

6.4 Clustering using K-Means

This figure shows the clustering results using the K-Means algorithm. Students are grouped based on similarity in attendance and marks. It helps in identifying performance patterns.

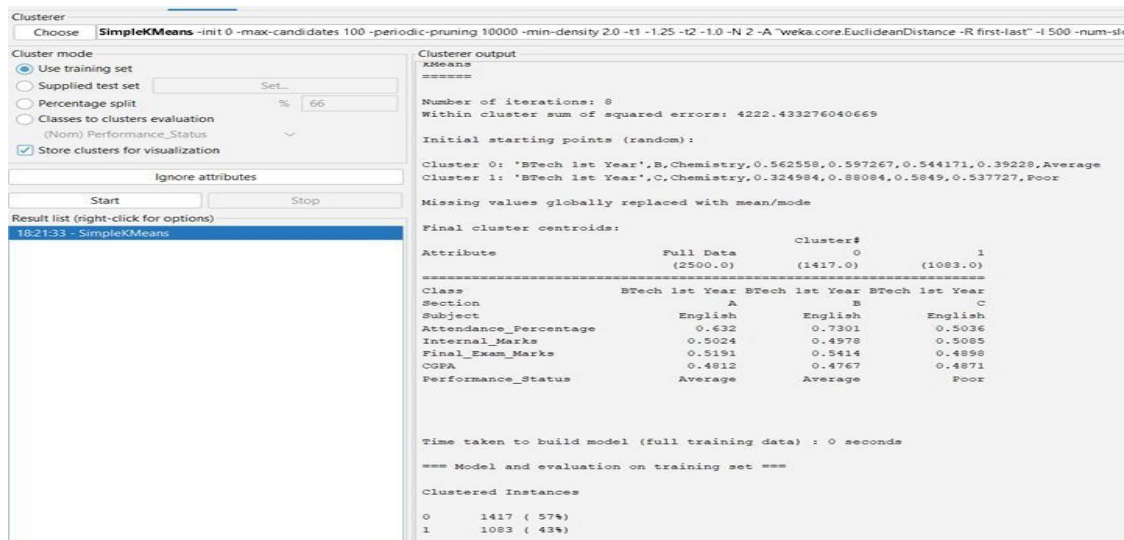


Figure 6.14 K-Means Clustering Output

6.5 Visualization of Results

This figure shows the relationship between attendance and CGPA. Each point represents a student in the dataset. It helps in identifying trends and patterns visually.

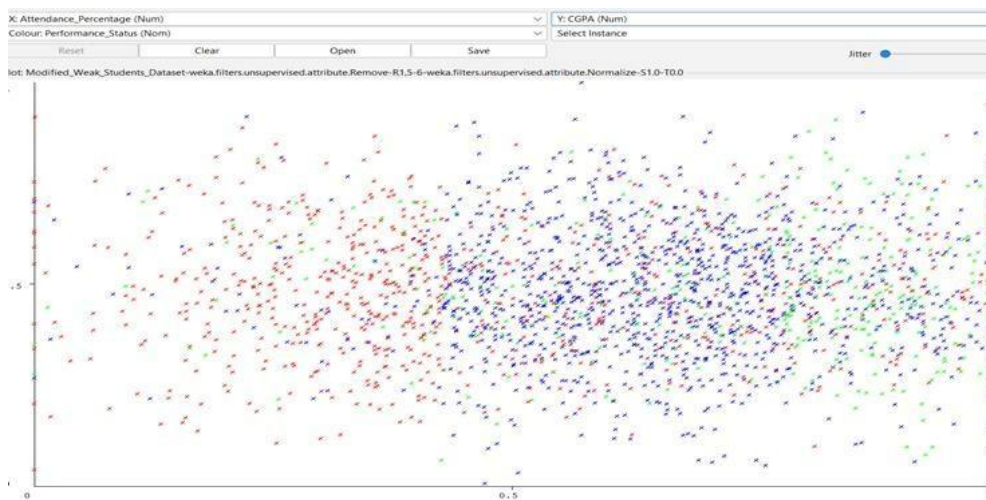


Figure 6.15: Scatter Plot Visualization

6.6 Association Rule Mining using Apriori Algorithm

This figure shows the Apriori algorithm used for association rule mining in WEKA. It generates rules based on attendance, marks, and performance using support and confidence values. The results show that higher attendance leads to better performance, while low attendance leads to poor results.

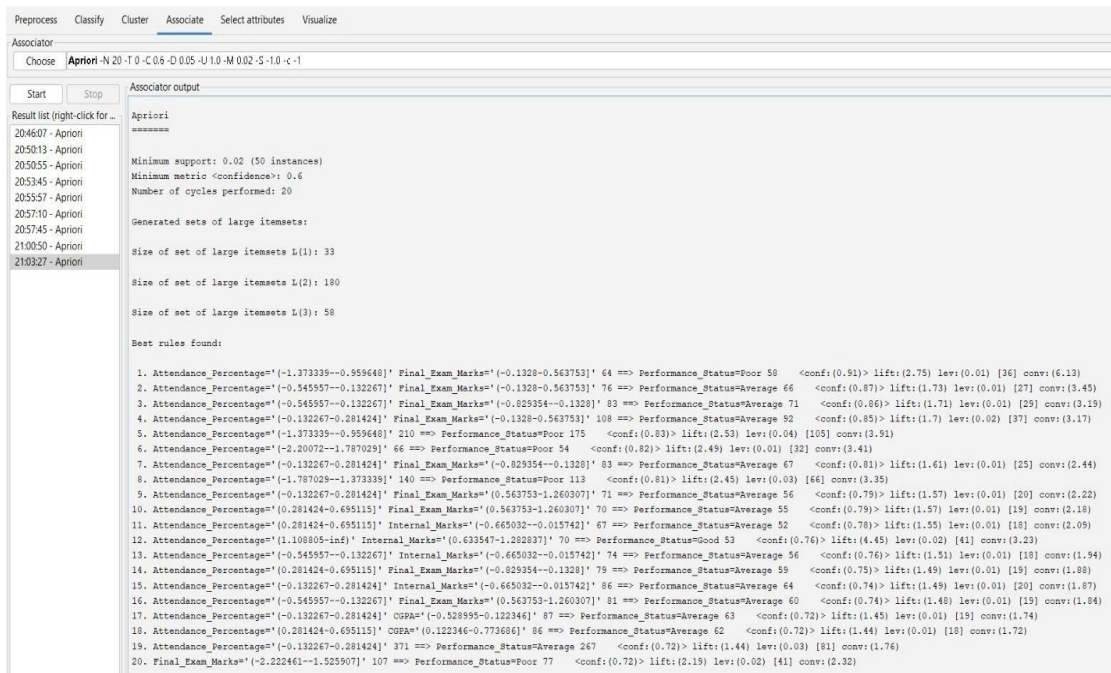


Figure 6.16: Apriori Association Rule Mining Output

7. RESULTS AND ANALYSIS

The results of the project show that attendance has a significant impact on student performance. The classification algorithm achieved an accuracy of approximately 79.72%, which indicates reliable performance. Students are categorized into three groups: Good, Average, and Poor. The analysis shows that students with higher attendance percentages tend to fall into the Good category, while those with lower attendance are often categorized as Poor.

Clustering results further support this observation by grouping students into clusters based on performance levels. Visualization using scatter plots clearly shows a positive relationship between attendance and CGPA.

Association rule mining using the Apriori algorithm further identifies strong relationships between attendance, marks, and performance, showing that low attendance is frequently linked to poor results while high attendance is associated with better performance.

These results confirm that attendance plays an important role in academic success.

8. CONCLUSION

The Attendance Analysis System successfully demonstrates how data mining techniques can be used to analyze student performance. The project highlights the importance of attendance in achieving better academic results.

By applying classification and clustering techniques, the system provides valuable insights into student behavior. It helps in identifying weak students at an early stage and supports educators in improving academic outcomes.

Overall, the project achieves its objectives and provides a useful tool for educational analysis.

9. FUTURE ENHANCEMENTS

The system can be further improved by adding advanced features. These include:

- Integration with real-time data systems
- Use of advanced machine learning algorithms
- Development of a web-based interface
- Addition of predictive analytics
- Inclusion of more attributes such as behavior and participation

These enhancements will make the system more powerful and suitable for real-world applications.

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